

Question:

A 66-year-old farmer is being evaluated for abnormal lung findings on a low dose chest CT scan obtained as part of his lung cancer screening. He has a 50-pack-year smoking history and has been hesitant to quit. He has a non-productive cough but brushes it away saying he is not bothered by it. He denies ever coughing up blood, breathlessness, chest pain, fatigue, or weight loss. He has never sought any medical care and states that he has always been in good shape. He consumes alcohol moderately and uses marijuana occasionally. He lives with his wife and has not traveled recently. On physical examination, his temperature is 37.1°C (98.8°F), blood pressure is 148/70 mm Hg, and pulse rate is 95/min. His BMI is 32 kg/m2. A general physical examination is unremarkable. Coarse breath sounds are present bilaterally. The cardiac exam is normal. Laboratory studies show a normal complete blood count and comprehensive metabolic panel. A follow-up high-resolution CT scan is performed that shows small irregular subcentimeter pulmonary nodules, several of which are cavitated in both lungs, predominantly distributed in the upper and middle zones. There is no mediastinal or hilar lymphadenopathy. A transbronchial needle aspiration of the lesion is performed which shows a nodular pattern of abundant, granular, mildly eosinophilic cells with grooved nuclei with indented nuclear membranes and a chronic inflammation that consists primarily of eosinophils. Immunohistochemical staining reveals numerous cells that stain positive for S100 and CD1a. Which of the cells of the human immune system are responsible for this lesion?

Answer Choices:

- A. T lymphocytes
- B. Natural killer cells
- C. Dendritic cells
- D. Ciliary epithelium

Reasoning + Answer:

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The clinical scenario describes a 66-year-old smoker with a 50-pack-year history, presenting with multiple small, irregular pulmonary nodules with cavitation on CT, and a biopsy showing granular, mildly eosinophilic cells with grooved nuclei, indented nuclear membranes, and chronic inflammation dominated by eosinophils. Immunohistochemical staining shows positive staining for S100 and CD1a.

Key features:

- S100 and CD1a are markers of **Langerhans cells**, which are a type of dendritic cell.
- Langerhans cell histiocytosis (LCH) is a rare disorder characterized by the proliferation of Langerhans cells, which are antigen-presenting cells of the dendritic cell lineage.
- The presence of granular cells with grooved nuclei, eosinophilic cytoplasm, and S100/CD1a positivity is classic for Langerhans cells.
- The nodules are subcentimeter, bilateral, with cavitation, and there is chronic eosinophilic inflammation consistent with Langerhans cell histiocytosis.
- The patient has a history of smoking, which is a known risk factor for LCH in some cases, though it is more common in younger patients. However, older age and chronic inflammation can still present with this entity.

Thus, the cells responsible for this lesion are Langerhans cells, which are a subset of dendritic cells.

Among the answer choices:

- A. T lymphocytes - incorrect; not associated with S100/CD1a
- B. Natural killer cells - incorrect; not positive for S100/CD1a
- C. Dendritic cells - correct; Langerhans cells are a type of dendritic cell
- D. Ciliary epithelium - incorrect; not involved in this histologic picture

Therefore, the correct answer is C. Dendritic cells.

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